

Distributed Computing
Class Test – I
Full Marks: 30

Answer all questions. All Questions have equal marks.

1. (CO1) In an implementation of Birman-Schiper-Stephenson's protocol for causal broadcast, checking for the second condition got left out by mistake. Show clearly with an example why the protocol will not work correctly in this system.
2. (CO2) Define the Chandy-Lamport snapshot algorithm. Mention all assumptions of the algorithm. Modify the algorithm to record channel states.
3. (CO3) Consider Ricart-Agarawala's mutual exclusion algorithm being run on a synchronous system with maximum message delay T for any link. Processing time at each node is negligible. The clocks are not synchronized but you can assume that they have zero drift. Is it possible to improve the message complexity (no. of messages per CS execution) of the algorithm by modifying it in some way? Justify your answer clearly.